

Development and Application of a Novel High-order Fully Actuated System Approach: Part II. 6-DOF Quadrotor Control

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Abstract—Traditional cascade controller design based on state-space modeling has been widely applied for quadrotor systems. The state-space representation can effectively model many dynamic systems. Yet, a group of first-order differential equations may not be the most suitable way to model and control inherent high-order dynamic systems, such as quadrotors. This paper proposes a modified high-order fully actuated (HOFA) approach with recursive actions based on a mixed-order quadrotor model. Unlike the existing HOFA approach developed for generalized strict-feedback systems, the modified HOFA method does not require the system in a strict-feedback form. Hence, the 6-DOF quadrotor model can be essentially converted into three HOFA subsystems. Based on the obtained HOFA systems, the control design of 6-DOF nonlinear quadrotor systems can be readily achieved like linear systems with arbitrarily assignable eigenstructure. Simulation and experimental results are shown to verify the effectiveness of the proposed HOFA modeling and control approach.

I. INTRODUCTION

In recent years, quadrotors have received significant attention for many different applications, such as surveillance, aerobatic maneuvers for entertainment, aerial photography, and transportation [1]-[3]. One significant advantage of quadrotors over other unmanned aerial vehicles (UAVs) is the ability to fly in any direction and perform vertical taking off/landing. A quadrotor is a highly coupled, under-actuated system with four rotor inputs and six outputs (three translations and three rotations). Generally, quadrotors are commanded to track three translations and the yaw rotation.

A hierarchical structure is commonly used for quadrotor control, which typically has two control loops. One is the inner control loop of attitude, and the other is the outer control loop of position. Linear control design, such as LQR and LQR-PID, was successfully applied to quadrotors due to their convenience and proven reliability [4]. To compensate partial nonlinear terms of the quadrotor model, nonlinear control techniques of quadrotor systems were introduced with strict mathematical proofs, such as feedback linearization, backstepping, sliding mode control, and robust adaptive tracking control [5]-[8]. People also investigated nonlinear manifolds, such as the quaternion [11][12] and the special Euclidean group [13], which avoided singularities and had less computational cost. To compensate complicated aero dynamics, the Lagrange formulation was applied to conveniently model gyroscopic effects and aerodynamic drag [10]. However, all these hierarchical controllers had

limitations on position tracking if the quadrotor could not closely track the desired attitude [14].

A different control design was presented based on a high-order dynamic model without assuming that the inner/attitude control was faster than the outer/position control [14]. The quadrotor was described by two second-order equations for altitude and yaw angle and two fourth-order equations for translational movements (x, y) . A high-order observer was designed to acquire thrust derivatives in the high-order system control. Although directly using high-order dynamics, the ad hoc modeling and control design of quadrotor systems was not systematically organized and generalizable for other inherent high-order dynamic systems.

A new system modeling and control method, the high order fully actuated (HOFA) approach, was recently developed to transfer nonlinear underactuated systems into linear systems via HOFA modeling [18]. Then, a globally stabilizable linear system was obtained with an arbitrarily assignable eigenstructure. The HOFA approach is more general and can be applied to nonlinear systems, for which the well-known backstepping and feedback linearization methods cannot be directly used [18]. Applying the HOFA approach to quadrotor systems will provide the advantages of direct control design without a hierarchical structure. Unfortunately, some nonlinear systems, such as quadrotors, cannot be converted into a strict-feedback form [15][16][20], which is a precondition of the HOFA approach [18]. The authors proposed a modified HOFA approach for a family of underactuated systems, including 3-DOF 2nd-order quadrotor systems, which did not require a strict-feedback form of the HOFA method [21]. Naturally, the planar motion can be extended to a 3D environment to achieve full control of the quadrotor. However, the method developed in [21] cannot be applied to 6-DOF quadrotor systems because mixed-order dynamic modeling and recursive actions [19] are required to use the HOFA approach.

Based on [19][21], this paper provides a modified HOFA approach with recursive actions, which can be applied to mixed-high-order dynamics systems. The modified HOFA method does not require each subsystem to be converted into a strict-feedback form [19]. As an example, a complete 6-DOF quadrotor system is divided into three high-order subsystems via the modified HOFA approach. After that, control inputs of each subsystem are readily obtained, and each closed-loop subsystem has arbitrarily assignable eigenstructures like linear systems.

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The rest of this paper is organized as follows. A modified HOFA approach with recursive actions for a mixed-order dynamic system is presented in Section II. The application of the modified HOFA approach to a 6-DOF quadrotor is presented in Section III. The simulation and experimental results are shown to verify the control performance in Section IV. Finally, we give the conclusion in Section V.

II. MODIFIED HOFA APPROACH WITH RECURSIVE ACTIONS

This section first introduces preliminaries of HOFA systems with recursive actions for mixed-order nonlinear systems [19][21]. For detailed descriptions, references [17]–[19][21] are suggested. Then, a modification of the HOFA approach for non-affined mixed-order systems is proposed.

Some commonly used symbols are defined as follows for $x \in \mathbb{R}^n$, and $A_i \in \mathbb{R}^{n \times n}$ ($i = 1, 2, \dots, m$),

$$\begin{aligned} x^{(0 \sim m)} &\triangleq \begin{bmatrix} x \\ \dot{x} \\ \vdots \\ x^{(m)} \end{bmatrix}, \\ x_{i \sim j}^{(0 \sim m)} &\triangleq \begin{bmatrix} x_i^{(0 \sim m)} \\ x_{i+1}^{(0 \sim m)} \\ \vdots \\ x_j^{(0 \sim m)} \end{bmatrix}, j \geq i, \\ A_{0 \sim m} &\triangleq [A_0 \quad A_1 \quad \dots \quad A_m]. \end{aligned} \quad (1)$$

A. Modeling and Control of HOFA Systems

Consider a common nonlinear system in an affine form,

$$\dot{x}^{(m)} = f(x^{(0 \sim m-1)}) + B(x^{(0 \sim m-1)})u, \quad (2)$$

where $x \in \mathbb{R}^n$ represents the state vector, $f(x^{(0 \sim m-1)}) \in \mathbb{R}^n$ is a differentiable vector function, $B(x^{(0 \sim m-1)}) \in \mathbb{R}^{n \times r}$ is a matrix function, and $u \in \mathbb{R}^r$ represents the input vector. With $\text{rank}(B(x^{(0 \sim m-1)})) = r = n$, (2) is a fully actuated system [18]. If $r > n$, (2) is an over-actuated system, and we can treat such systems as fully actuated ones similarly.

Proposition 1 [18]: Denoting a set of matrices $A_i \in \mathbb{R}^{r \times r}$, $i = 0, 1, \dots, m-1$ presents the following controller

$$\begin{cases} u = -B^{-1}(x^{(0 \sim m-1)})(A_{0 \sim m-1}x^{(0 \sim m-1)} + u^*) \\ u^* = f(x^{(0 \sim m-1)}) - v \end{cases}. \quad (3)$$

Then the fully actuated system (2) can be rewritten as the following linear closed-loop system,

$$\dot{x}^{(m)} + A_{0 \sim m-1}x^{(0 \sim m-1)} = v. \quad (4)$$

We can select appropriate A_i and external signal v [18] to acquire arbitrary eigenvalues or eigenstructure of the linear system (4). This is an ideal property of the HOFA modeling for general nonlinear systems control design. Many nonlinear systems can be converted into HOFA systems [18].

B. A Modified HOFA Approach for Mixed-Order Nonlinear Systems with Recursive Actions

The author's previous work [21] discussed a modified HOFA method for a 3-DOF quadrotor (2nd order model) for a non-strict-feedback form but with unitary orders. Unlike the 3-DOF quadrotor model, the 6-DOF quadrotor model and many complex nonlinear systems contain mixed-order terms.

Based on [19], we present the modified HOFA approach with recursive actions for the 6-DOF quadrotor model.

Considering a mixed-order nonlinear system [19],

$$\begin{bmatrix} \dot{x}_1^{(m_1)} \\ \vdots \\ \dot{x}_n^{(m_n)} \end{bmatrix} = f(\cdot) + B(\cdot)u, \quad (5)$$

where $m_1, \dots, m_n$ are a set of positive integers. To apply the HOFA approach, a unimodular matrix Q with $z = Qx$, ($z = [z_1, z_2, \dots, z_p]^T \in \mathbb{R}^n$), is found to convert (5) into p subsystems. Denote the i th subsystem with $z_i = [z_{i,1}, z_{i,2}, \dots, z_{i,n_i}]^T$, $z_{i,j} \in \mathbb{R}^{r_i}$, $j \in [1 \ n_i]$, $u_i \in \mathbb{R}^{r_i}$ with $\sum_{i=1}^p r_i = r$, the i th subsystem is written as,

$$\begin{cases} \dot{z}_{i,1}^{(m_{i,1})} = f_{i,1}(z_{i,1}^{(0 \sim m_{i,1}-1)}) + g_{i,1}(z_{i,1}^{(0 \sim m_{i,1}-1)}, z_{i,2}), \\ \dot{z}_{i,2}^{(m_{i,2})} = f_{i,2}(z_{i,1}^{(0 \sim m_{i,1}-1)}, z_{i,2}^{(0 \sim m_{i,2}-1)}) \\ \quad + g_{i,2}(z_{i,1}^{(0 \sim m_{i,1}-1)}, z_{i,2}^{(0 \sim m_{i,2}-1)}, z_{i,3}), \\ \vdots \\ \dot{z}_{i,n_i}^{(m_{i,n_i})} = f_{i,n_i}(z_{i,j}^{(0 \sim m_{i,j}-1)}|_{j=1 \sim n_i-1}, z_{i,n_i}^{(0 \sim m_{i,n_i}-1)}) \\ \quad + g_{i,n_i}(z_{i,j}^{(0 \sim m_{i,j}-1)}|_{j=1 \sim n_i-1}, z_{i,n_i}^{(0 \sim m_{i,n_i}-1)})u_i. \end{cases} \quad (6)$$

To substitute $z_{i,2}^{(m_{i,2})}$ in the 2nd equation of (6) back to the 1st equation, the 1st equation (also $g_{i,1}$) will need to take the $m_{i,2}$ th derivative. In this process, generally, the $m_{i,j+1}$ th order derivative ($j \in [1 \ n_i - 1]$) of $g_{i,j}$ can be written as,

$$\begin{aligned} &g_{i,j}^{(m_{i,j+1})}(z_{i,k}^{(0 \sim m_{i,k}-1)}|_{k=1 \sim j-1}, z_{i,j}^{(0 \sim m_{i,j}-1)}, z_{i,j+1}) \\ &= G_{i,j}(z_{i,k}^{(0 \sim m_{i,k}+m_{i,j+1})}|_{k=1 \sim j-1}, z_{i,j}^{(0 \sim m_{i,j}+m_{i,j+1}-1)}), \\ &\quad z_{i,j+1}^{(0 \sim m_{i,j+1}-1)}) + \\ &H_{i,j}(z_{i,k}^{(0 \sim m_{i,k}-1)}|_{k=1 \sim j-1}, z_{i,j}^{(0 \sim m_{i,j}-1)}, z_{i,j+1})z_{i,j+1}^{(m_{i,j+1})}. \end{aligned} \quad (7)$$

Remark 1: Different from the mixed-order strict-feedback nonlinear system in [19], all p subsystems in the form of (6) do not require strict-feedback forms. High order terms $z_{i,j+1}^{(m_{i,j+1})}$, $j \in [1 \ n_i - 1]$ can always be separated from $g_{i,j}^{(m_{i,j+1})}$.

Remark 2: For some highly coupled systems, such as quadrotors, we cannot allocate states and inputs to each subsystem strictly by finding the unimodular matrix Q [21]. For each subsystem, it is acceptable to introduce singularity at specific points and to have $f_{i,j}$, $g_{i,j}$ matrices containing states and inputs of other subsystems.

Assumption 1: The coefficient matrices $H_{i,j} \in \mathbb{R}^{r_i \times r_i}$, $i = 1, 2, \dots, p$, $j = 1, 2, \dots, n_i - 1$ are nonsingular in most cases.

$$\det H_{i,j}(z_{i,k}^{(0 \sim m_{i,k}-1)}|_{k=1 \sim j-1}, z_{i,j}^{(0 \sim m_{i,j}-1)}, z_{i,j+1}) \neq 0. \quad (8)$$

We also require the coefficient before the control for each subsystem to be non-zero.

$$\det g_{i,n_i}(z_{i,j}^{(0 \sim m_{i,j}-1)}|_{j=1 \sim n_i-1}, z_{i,n_i}^{(0 \sim m_{i,n_i}-1)}) \neq 0. \quad (9)$$

Introducing the notation,

$$q_{i,1 \sim k} = \sum_{j=1}^k m_{i,j}, \quad i = 1, \dots, p, \quad (10)$$

we have the following Theorem.

Theorem 1: Let Assumption 1 hold. Then, each subsystem (6) can be transformed into the following HOFA system,

$$\begin{aligned} z_{i,1}^{(q_{i,1 \sim n_i})} &= L_i \left(z_i^{(0 \sim q_{i,1 \sim n_i} - 1)} \right) + \\ &H_i \left(z_i^{(0 \sim q_{i,1 \sim n_i} - 1)} \right) u_i. \end{aligned} \quad (11)$$

Proof: Considering $n_i > 2$, we first take $m_{i,2}$ th-order derivative of $z_{i,1}^{(m_{i,1})}$

$$\begin{aligned} z_{i,1}^{(m_{i,1} + m_{i,2})} &= z_{i,1}^{(q_{i,1 \sim 2})} = f_{i,1}^{(m_{i,2})} \left(z_{i,1}^{(0 \sim m_{i,1} - 1)} \right) + \\ g_{i,1}^{(m_{i,2})} \left(z_{i,1}^{(0 \sim m_{i,1} - 1)}, z_{i,2} \right) &= F_{i,1} \left(z_{i,1}^{(0 \sim q_{i,1 \sim 2} - 1)} \right) + \\ G_{i,1} \left(z_{i,1}^{(0 \sim q_{i,1 \sim 2} - 1)}, z_{i,2}^{(0 \sim m_{i,2} - 1)} \right) &+ \\ H_{i,1} \left(z_{i,1}^{(0 \sim m_{i,1} - 1)} \right) z_{i,2}^{(m_{i,2})} & \\ = L_{i,1}(\cdot) + H_{i,1}(\cdot) z_{i,2}^{(m_{i,2})}, \end{aligned} \quad (12)$$

where

$$L_{i,1}(\cdot) = F_{i,1}(\cdot) + G_{i,1}(\cdot), \quad (13)$$

and substitute expression of $z_{i,2}^{(m_{i,2})}$ from (6)

$$\begin{aligned} z_{i,1}^{(q_{i,1 \sim 2})} &= L_{i,1}(\cdot) + H_{i,1}(\cdot) (f_{i,2}(\cdot) + \\ g_{i,2}(\cdot)) &= l_{i,1}(\cdot) + H_{i,1}(\cdot) G_{i,2}(\cdot), \end{aligned} \quad (14)$$

where

$$l_{i,1}(\cdot) = L_{i,1}(\cdot) + H_{i,1}(\cdot) f_{i,2}(\cdot). \quad (15)$$

Secondly, we take $m_{i,3}$ th-order derivative of (14) to have

$$\begin{aligned} z_{i,1}^{(q_{i,1 \sim 3})} &= l_{i,1}^{(m_{i,3})}(\cdot) + \sum_{k=0}^{m_{i,3}-1} H_{i,1}^{(m_{i,3}-k)} g_{i,2}^{(k)} + \\ &H_{i,1}^{(0)} g_{i,2}^{(m_{i,3})}. \end{aligned} \quad (16)$$

According to (7), we can have the last term in (16) as

$$H_{i,1} g_{i,2}^{(m_{i,3})} = H_{i,1}(\cdot) G_{i,2}(\cdot) + H_{i,1}(\cdot) H_{i,2}(\cdot) z_{i,3}^{(m_{i,3})}. \quad (17)$$

Now we can rewrite (16) as

$$z_{i,1}^{(q_{i,1 \sim 3})} = L_{i,2}(\cdot) + H_{i,1}(\cdot) H_{i,2}(\cdot) z_{i,3}^{(m_{i,3})}, \quad (18)$$

where

$$\begin{aligned} L_{i,2}(\cdot) &= l_{i,1}^{(m_{i,3})}(\cdot) + \sum_{k=0}^{m_{i,3}-1} H_{i,1}^{(m_{i,3}-k)} g_{i,2}^{(k)} + \\ &H_{i,1}(\cdot) G_{i,2}(\cdot). \end{aligned} \quad (19)$$

Notice the last terms in (12)(18), it is trivial to have the expression $z_{i,1}^{(q_{i,1 \sim (j+1)})}$ at j th iteration step ($1 < j < n_i - 1$)

$$z_{i,1}^{(q_{i,1 \sim (j+1)})} = L_{i,j}(\cdot) + \prod_{k=1}^j H_{i,k}(\cdot) z_{i,j+1}^{(m_{i,j+1})} \quad (20)$$

Thus, we have the expression of $z_{i,1}^{(q_{i,1 \sim n_i})}$

$$\begin{aligned} z_{i,1}^{(q_{i,1 \sim n_i})} &= L_{i,n_i-1}(\cdot) + \prod_{k=1}^{n_i-1} H_{i,k}(\cdot) z_{i,n_i}^{(m_{i,n_i})} \\ &= L_{i,n_i-1}(\cdot) + \prod_{k=1}^{n_i-1} H_{i,k}(\cdot) (f_{i,n_i}(\cdot) + g_{i,n_i} u_i) \end{aligned} \quad (21)$$

As (8) and (9) hold, (21) is an effective HOFA system. If $n_i = 1$, (11) is already an HOFA system. If $n_i = 2$, we just need skip the steps (14)-(19). The proof is complete. ■

Based on Theorem 1, we can apply the new HOFA method with recursive actions to the quadrotor control with 6-DOF.

III. HOFA APPLICATIONS: 6-DOF QUADROTOR CONTROL

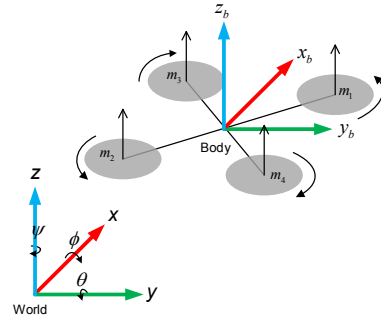


Figure 1. 6-DOF quadrotor.

A 6-DOF quadrotor system is shown in Figure 1. Let $[x, y, z]^T \in \mathbb{R}^3$ denote positions with respect to the inertial frame, $[\phi, \theta, \psi]^T \in \mathbb{R}^3$ denote roll, pitch and yaw angles to the inertial frame, and $[p, q, r]^T \in \mathbb{R}^3$ denote angular velocities of roll, pitch, and yaw to the body-fixed frame. The mixed-order nonlinear model is given below,

$$\begin{cases} \ddot{x} = \frac{T}{m} (-c\psi s\theta c\phi - s\phi c\psi) \\ \ddot{y} = \frac{T}{m} (c\psi s\phi - s\psi s\theta c\phi) \\ \ddot{z} = \frac{T}{m} c\phi c\theta - g \\ \dot{\phi} = p + q s\phi t\theta - r c\phi t\theta \\ \dot{\theta} = q c\phi + r s\phi \\ \dot{\psi} = -q \frac{s\phi}{c\theta} + r \frac{c\phi}{c\theta} \\ \dot{p} = \frac{J_y - J_z}{J_x} q r + \frac{\tau_\phi}{J_x} \\ \dot{q} = \frac{J_z - J_x}{J_y} p r + \frac{\tau_\theta}{J_y} \\ \dot{r} = \frac{J_x - J_y}{J_z} p q + \frac{\tau_\psi}{J_z} \end{cases}, \quad (22)$$

where c denotes the cosine function, s denotes the sine function, and t represents the tangent function. T is the total

rotor thrust, $[\tau_\phi, \tau_\theta, \tau_\psi]^T \in \mathbb{R}^3$ denotes the body torques, m is the quadrotor mass, $[J_x, J_y, J_z]^T \in \mathbb{R}^3$ denotes the moment of inertia to the body frame, and g is the gravity acceleration. To transfer (22) to a HOFA system, we first select the Q matrix,

$$Q = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad (23)$$

and denotes $z_1 = z$, $z_{2,1} = [x, y]^T$, $z_{2,2} = [\phi, \theta]^T$, $z_{2,3} = [p, q]^T$, $z_{3,1} = \psi$, and $z_{3,2} = r$. The control inputs are defined as follows,

$$u_1 = \frac{T}{m}, u_2 = \begin{bmatrix} \tau_\phi/J_x \\ \tau_\theta/J_y \end{bmatrix}, u_3 = \frac{\tau_\psi}{J_z}. \quad (24)$$

To write the following derivation concisely, we also define the following terms,

$$\begin{aligned} g_1 &= c\phi c\theta, \\ g_{2,1} &= \begin{bmatrix} -s\phi & -s\theta c\phi \\ -s\theta c\phi & s\phi \end{bmatrix}, \\ h_{2,1} &= \begin{bmatrix} s\psi \\ c\psi \end{bmatrix}, f_{2,2} = \begin{bmatrix} -r c\phi t\theta \\ r s\phi \end{bmatrix}, \\ g_{2,2} &= \begin{bmatrix} 1 & s\phi t\theta \\ 0 & c\phi \end{bmatrix}, f_{2,3} = \begin{bmatrix} \frac{J_y - J_z}{J_x} qr \\ \frac{J_z - J_x}{J_y} pr \end{bmatrix}, \\ f_{3,1} &= -q \frac{s\phi}{c\theta}, g_{3,1} = \frac{c\phi}{c\theta}, f_{3,2} = \frac{J_x - J_y}{J_z} pq. \end{aligned} \quad (25)$$

Then the mixed-order nonlinear 6-DOF quadrotor system (22) is transferred to three subsystems, and each subsystem is in the form of (6),

$$z_1^{(2)} = g_1 u_1 - g, \quad (26)$$

$$\begin{cases} z_{2,1}^{(2)} = g_{2,1} h_{2,1} u_1 \\ z_{2,2}^{(1)} = f_{2,2} + g_{2,2} z_{2,3}, \\ z_{2,3}^{(1)} = f_{2,3} + u_2 \end{cases} \quad (27)$$

$$\begin{cases} z_{3,1}^{(1)} = f_{3,1} + g_{3,1} z_{3,2} \\ z_{3,2}^{(1)} = f_{3,2} + u_3 \end{cases}. \quad (28)$$

Note that the altitude in (26), the translational position x, y in (27), and the yaw angle in (28) are the desired states to be controlled. $A_{0 \sim 1}^{z_1} = [16, 8]$, $v_1 = [16, 4] z_{1, \text{ref}}^{(0 \sim 1)}$

First, (26) is already a HOFA system. The controller can be immediately obtained,

$$u_1 = g_1^{-1} (g - A_{0 \sim 1}^{z_1} z_1^{(0 \sim 1)} + v_1), \quad (29)$$

where v_1 is a designed external signal and $A_{0 \sim 1}^{z_1}$ is selected to make the closed-loop system have the desired eigenstructure,

$$z_1^{(2)} + A_{0 \sim 1}^{z_1} z_1^{(0 \sim 1)} = v_1. \quad (30)$$

Second, to get the HOFA model of (27), taking the first-order derivative of $z_{2,1}^{(2)}$ will give,

$$z_{2,1}^{(3)} = u_1 g_{2,1}^{(1)} h_{2,1} + u_1 g_{2,1} h_{2,1}^{(1)} + u_1^{(1)} g_{2,1} h_{2,1}, \quad (31)$$

with

$$\begin{aligned} g_{2,1}^{(1)} &= \begin{bmatrix} -\dot{\phi} c\phi & \delta_1(\cdot) \\ \delta_1(\cdot) & \dot{\phi} c\phi \end{bmatrix}, \\ \delta_1(\cdot) &= -\dot{\theta} c\theta c\phi + \dot{\phi} s\theta s\phi, \\ h_{2,1}^{(1)} &= \begin{bmatrix} \dot{\psi} c\psi \\ -\dot{\psi} s\psi \end{bmatrix}. \end{aligned} \quad (32)$$

Based on the definition of $z_{2,2} = [\phi, \theta]^T$, $u_1 g_{2,1}^{(1)} h_{2,1}$ in (31) can be rewritten by substituting (32) in,

$$u_1 g_{2,1}^{(1)} h_{2,1} = u_1 \underbrace{\begin{bmatrix} s\theta s\phi c\psi - c\phi s\psi & -c\phi c\theta c\psi \\ s\theta s\phi s\psi + c\phi c\psi & -c\phi c\theta s\psi \end{bmatrix}}_{H_{2,1}} z_{2,2}^{(1)}. \quad (33)$$

Let

$$G_{2,1} = u_1 g_{2,1} h_{2,1}^{(1)} + u_1^{(1)} g_{2,1} h_{2,1}, \quad (34)$$

we have

$$z_{2,1}^{(3)} = G_{2,1} + H_{2,1} z_{2,2}^{(1)}, \quad (35)$$

which is in the form of (12). Substituting the expression of $z_{2,2}^{(1)}$ in (27) into (35) gives,

$$z_{2,1}^{(3)} = G_{2,1} + H_{2,1} f_{2,2} + H_{2,1} g_{2,2} z_{2,3}. \quad (36)$$

Then, we retake the first-order derivative for (36),

$$\begin{aligned} z_{2,1}^{(4)} &= G_{2,1}^{(1)} + H_{2,1}^{(1)} f_{2,2} + H_{2,1} f_{2,2}^{(1)} + \\ &H_{2,1}^{(1)} g_{2,2} z_{2,3} + H_{2,1} g_{2,2}^{(1)} z_{2,3} + H_{2,1} g_{2,2} z_{2,3}^{(1)}, \end{aligned} \quad (37)$$

which is in the form of (18). Substituting the expression of $z_{2,3}^{(1)}$ in (27) into (37) gives,

$$\begin{aligned} z_{2,1}^{(4)} &= G_{2,1}^{(1)} + H_{2,1}^{(1)} f_{2,2} + H_{2,1} f_{2,2}^{(1)} + H_{2,1}^{(1)} g_{2,2} z_{2,3} \\ &+ H_{2,1} g_{2,2}^{(1)} z_{2,3} + H_{2,1} g_{2,2} (f_{2,3} + u_2), \end{aligned} \quad (38)$$

which is in the form of (21). Denote

$$\begin{aligned} l_{2,2} &= G_{2,1}^{(1)} + H_{2,1}^{(1)} f_{2,2} + H_{2,1} f_{2,2}^{(1)} + H_{2,1}^{(1)} g_{2,2} z_{2,3} \\ &+ H_{2,1} g_{2,2}^{(1)} z_{2,3} + H_{2,1} g_{2,2} f_{2,3}, \end{aligned} \quad (39)$$

and the controller can be readily obtained for the HOFA system (38),

$$u_2 = (H_{2,1} g_{2,2})^{-1} (-l_{2,2} - A_{0 \sim 3}^{z_{2,1}} z_{2,1}^{(0 \sim 3)} + v_2), \quad (40)$$

where v_2 is a designed external signal and $A_{0 \sim 3}^{z_{2,1}}$ is selected to make the closed-loop system have desired eigenstructure,

$$z_{2,1}^{(4)} + A_{0 \sim 3}^{z_{2,1}} z_{2,1}^{(0 \sim 3)} = v_2. \quad (41)$$

Remark 3: The control in (40) contains the high-order terms $z_{2,1}^{(3)}$ and derivatives of the input u_1 (check $G_{2,1}^{(1)}$ and $H_{2,1}^{(1)}$ in $l_{2,2}$), which may need to be estimated smoothly.

Finally, to get the HOFA model of (28), taking the first-order derivative of $z_{3,1}^{(1)}$ will give,

$$z_{3,1}^{(2)} = f_{3,1}^{(1)} + g_{3,1}^{(1)} z_{3,2} + g_{3,1} z_{3,2}^{(1)}. \quad (42)$$

Substituting $z_{3,2}^{(1)}$ of (28) into (42), the following HOFA system is obtained in the form of (21),

$$z_{3,1}^{(2)} = f_{3,1}^{(1)} + g_{3,1}^{(1)} z_{3,2} + g_{3,1} (f_{3,2} + u_3). \quad (43)$$

and again, the controller is readily obtained,

$$u_3 = -f_{3,2} - g_{3,1}^{-1} \left(f_{3,1}^{(1)} + g_{3,1}^{(1)} z_{3,2} + A_{0 \sim 1}^{z_{3,1}} z_{3,1}^{(0 \sim 1)} - v_3 \right), \quad (44)$$

where v_3 is a designed external signal and $A_{0 \sim 1}^{z_{3,1}}$ is selected to make the closed-loop system have desired eigenstructure,

$$z_{3,1}^{(2)} + A_{0 \sim 1}^{z_{3,1}} z_{3,1}^{(0 \sim 1)} = v_3. \quad (45)$$

Remark 4: to avoid singularity in (29), (40), and (44), constraints at near-zero intervals are needed for u_1 , $\cos \phi$, and $\cos \theta$, which are trivial and work for large-angle flying maneuvers [22].

IV. SIMULATION AND EXPERIMENT RESULTS

This section presents simulation results in MATLAB and experimental results to validate the proposed modified HOFA approach to control a 6-DOF quadrotor modeled in a mixed-order nonlinear system. The quadrotor parameters applied in simulation and experiments are shown in Table 1.

Table 1. Model parameters of the quadrotor

Symbol	Definition	Value
m	mass	0.625 kg
g	Acceleration due to gravity	9.8 m/s ²
J_x	Moment of inertia in the X axis	0.0019005 kg · m ²
J_y	Moment of inertia in the Y axis	0.0019536 kg · m ²
J_z	Moment of inertia in the Z axis	0.0036894 kg · m ²

A. Simulation Results and Discussions

Simulation results with two sets of control parameters demonstrate that the obtained linear systems from the modified HOFA method can be assigned arbitrary eigenstructure.

Control Parameter Set 1:

$$\begin{aligned} A_{0 \sim 1}^{z_1} &= [16, 8], v_1 = [16, 4] z_{1,\text{ref}}^{(0 \sim 1)}, \\ A_{0 \sim 3}^{z_{2,1}} &= \begin{bmatrix} 256, 256, 96, 16, 0_{1 \times 4} \\ 0_{1 \times 4}, 256, 256, 96, 16 \end{bmatrix}, \\ v_2 &= [256, 144, 4] z_{2,1,\text{ref}}^{(0 \sim 2)}, \\ A_{0 \sim 1}^{z_{3,1}} &= [16, 8], v_3 = [16, 4] z_{3,1,\text{ref}}^{(0 \sim 1)}. \end{aligned} \quad (46)$$

The closed-loop system can also be written in the frequency domain with a capital Z as the state,

$$\begin{aligned} Z_1 &= K(s) Z_{1,\text{ref}}, \\ Z_{2,1} &= K(s) I_{2 \times 2} Z_{2,1,\text{ref}}, \\ Z_{3,1} &= K(s) Z_{3,1,\text{ref}}, K(s) = \frac{4}{s + 4}. \end{aligned} \quad (47)$$

Control Parameter Set 2 (faster responses):

$$\begin{aligned} A_{0 \sim 1}^{z_1} &= [25, 10], v_1 = [25, 9] z_{1,\text{ref}}^{(0 \sim 1)}, \\ A_{0 \sim 3}^{z_{2,1}} &= \begin{bmatrix} 625, 500, 150, 20, 0_{1 \times 4} \\ 0_{1 \times 4}, 625, 500, 150, 20 \end{bmatrix}, \\ v_2 &= [625, 500, 100] z_{2,1,\text{ref}}^{(0 \sim 2)}, \\ A_{0 \sim 1}^{z_{3,1}} &= [25, 10], v_3 = [25, 9] z_{3,1,\text{ref}}^{(0 \sim 1)}. \end{aligned} \quad (48)$$

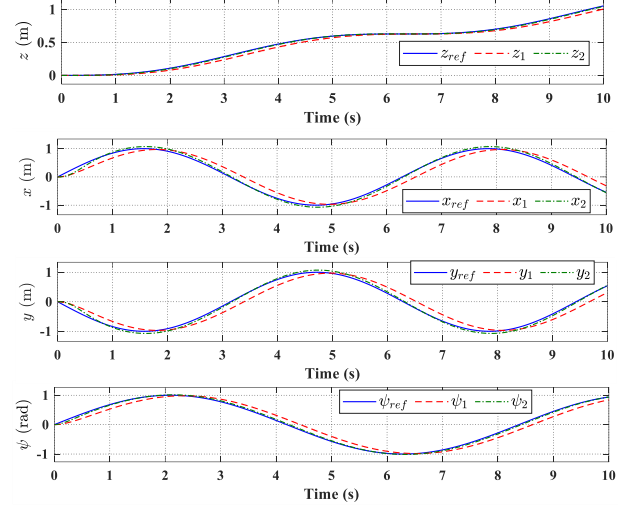


Figure 2. Tracking results of two different sets of control parameters.

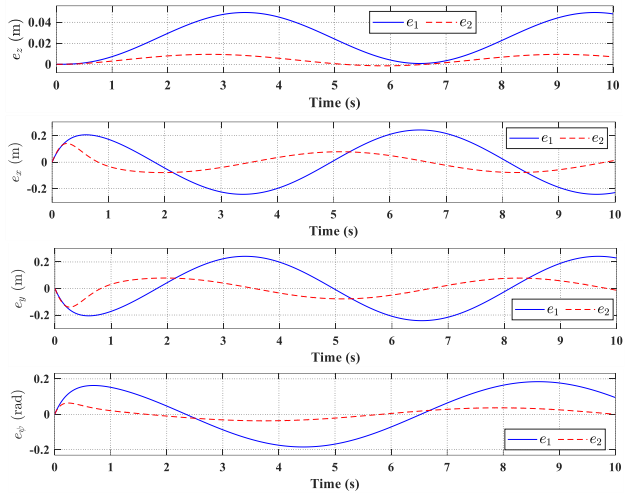


Figure 3. Tracking errors of two different sets of control parameters.

The closed-loop system can also be written in the frequency domain,

$$\begin{aligned} Z_1 &= K_1(s) Z_{1,\text{ref}}, K_1(s) = \frac{9s + 25}{s^2 + 10s + 25}, \\ Z_{2,1} &= K_2(s) I_{2 \times 2} Z_{2,1,\text{ref}}, \\ K_2(s) &= \frac{100s^2 + 500s + 625}{s^4 + 20s^3 + 150s^2 + 500s + 625}, \\ Z_{3,1} &= K_1(s) Z_{3,1,\text{ref}}. \end{aligned} \quad (49)$$

The double-integrated sinusoidal signal is applied as the reference input of altitude, and sinusoidal signals as the reference inputs of translational positions and yaw angle. The whole trajectory depicts a Lemniscate shape widely used for quadrotor demonstrations. Simulation results are presented in Figure 2 and Figure 3. The second set has a quicker response

with smaller tracking errors. The result indicates the significant advantage that the HOFA controller can easily achieve different control performances by arbitrarily assigning different eigenstructures, which is easy tuning and characteristics of linear systems.

B. Experimental Results and Discussions

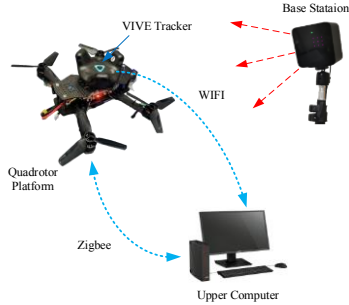


Figure 4. A 6-DOF quadrotor experimental system.

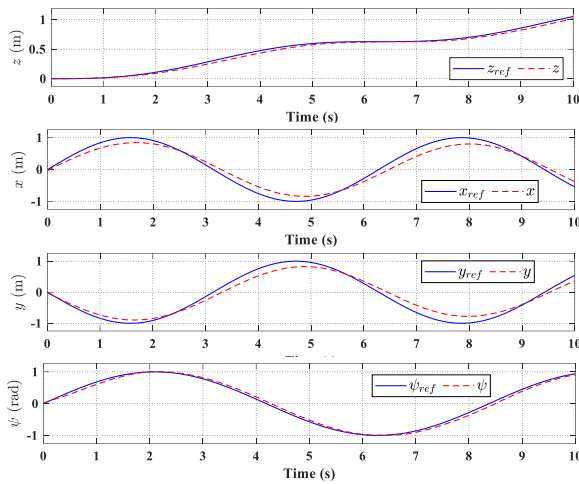


Figure 5. Experimental tracking results.

The quadrotor system shown in Figure 4 is used to verify the HOFA controller. The Vive tracker passively estimates the tracking data by sensing the base station's light. The Vive tracker provides position and attitude tracking data to the upper computer. Commanding and tracking messages are exchanged between the upper computer and the quadrotor flight controller via the ZigBee network. The experimental tracking results are shown in Figure 5. Considering safety and actuator saturation, we chose the first set of control parameters. The plots show the effectiveness of the HOFA approach applied to the quadrotor platform.

V. CONCLUSION

This paper presented a modified HOFA approach with recursive actions for mixed-order nonlinear systems, which does not require a strict feedback form. The new approach is applied to a 6-DOF quadrotor model with simulation and experimental results. Future works include high-order terms estimation in the HOFA approach.

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